American Computer Science League

2020-2021 ● Contest 4: Shorts ● Intermediate Division

1. Graph Theory

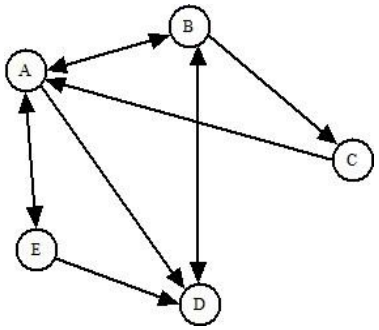
Given the following adjacency matrix, which vertices do not have a path of length 3 to vertex B?

$$\begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{array}$$

A. A, C, D
B. A, B, C
C. A, B, D
D. C, D
E. A, C

2. Graph Theory

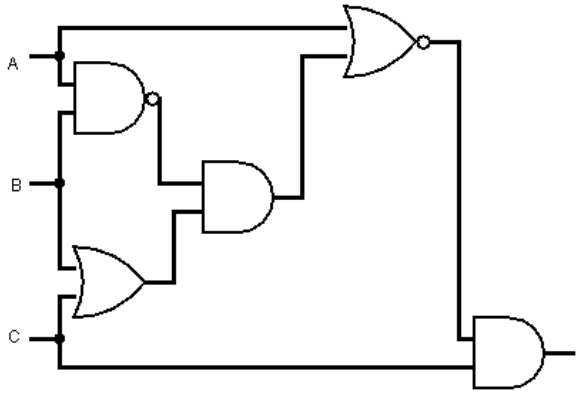
How many cycles are there from vertex B in this directed graph?



A. 5
B. 6
C. 7
D. 8
E. 9

3. Digital Electronics

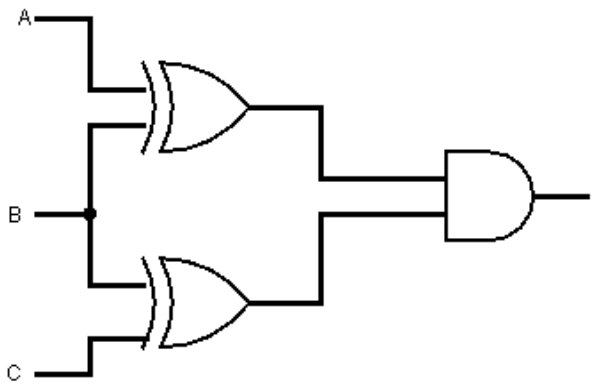
How many ordered triples make this circuit TRUE?



A. 0
B. 1
C. 2
D. 3
E. 4

4. Digital Electronics

Identify the ordered triples that make this circuit TRUE:



A. (0,*,0)
B. (1,*,1)
C. (0,*,0), (1,0,1)
D. (1,*,1), (0,1,0)
E. (0,1,0), (1,0,1)

5. Assembly Language

What is printed when the following program is executed?

```
X     DC      21
Y     DC      49
TOP   BE      STOP
      LOAD    X
      DIV     Y
      MULT    Y
      STORE   R
      LOAD    X
      SUB     R
      STORE   R
      LOAD    Y
      STORE   X
      LOAD    R
      STORE   Y
      BU      TOP
STOP  PRINT   X
      END
```

A. 1
B. 7
C. 14
D. 21
E. 49